blood

721. CLINICAL ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TRANSPLANT TOXICITIES: MICROBIODATA, ENDOTHELIAL DAMAGE, AND OPPORTUNISTIC INFECTIONS | NOVEMBER 29, 2018

Multicenter Microbiota Analysis Indicates That Pre-HCT Microbiota Injury Is Prevalent across Geography and Predicts Poor Overall Survival

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Abstract

Intestinal microbiota composition is associated with important outcomes after allo-HCT including survival, relapse, GVHD, and infections. These observations have been made almost exclusively by characterizing the microbiota in the first weeks after transplantation, and in single-center studies. We previously reported that intestinal diversity measured peri-neutrophil engraftment is predictive of overall survival in a multicenter cohort. Here, we hypothesized that pre-HCT microbiota configuration may also be an important determinant of post-transplantation outcomes. We report a multicenter analysis conducted at 4 independent international institutions to test this hypothesis.

We collected 1922 stool samples ~weekly from 991 unique allo-HCT patients at four international transplant centers: Cohorts 1 and 2 in the US, cohort 3 in Europe, and cohort 4 in Japan. The patients-all adult recipients of allo-HCT-varied in underlying diagnosis, donor-graft sources, conditioning intensity, and GVHD prophylaxis. Samples from all 4 centers were sequenced and analyzed at a central laboratory using the V4-V5 region of 16S rRNA. For patients with multiple samples within the pre-HCT sampling period, which we defined as day -20 to 0, median values were used.

On average, patients from all 4 transplantation centers had reduced microbiota diversity pre-HCT, as measured by median α-diversity (inverse Simpson) values that were 1.7-to-2.5-fold lower than those of healthy volunteers. This comparison was made both in volunteers whose samples we sequenced ourselves and in a publicly available dataset (**Fig A**, $p<0.005$). Since α-diversity measurements do not consider which taxa are present in a community, we asked whether the composition of pre-HCT microbiotas are similar to healthy microbial communities. While the intestinal communities of most healthy volunteers could be matched to the Enterotypes classifier, pre-HCT samples from all four centers had configurations that were poorly characterized by this independent classification scheme (**Fig B**). Thus, the post-HCT microbiota injuries that we previously observed comparably across geography are preceded by community structures that are already abnormal pre-HCT, consistent with our prior observation that pre-HCT antibiotic exposure is a risk factor for poor outcomes.

We next asked how similar these pre-HCT communities are across geography. We found that Bray-Curtis distances between institutions were reproducibly much smaller in magnitude than the changes observed over time during transplantation (**Fig C**, $p<0.005$). We also observed in the largest cohort (#1) that pre-HCT diversity is associated with patient survival. Among 753 patients, those in the lowest quartile of pre-HCT α-diversity had a lower overall survival than those in the highest quartile (**Fig D**, $p<0.009$).

In order to characterize these clinically relevant low-diversity phenotypes, we defined domination as a microbiota injury in which any operational taxonomic unit comprises >30% of bacterial abundance. The dominating taxa belonged to multiple genera, the most common being *Enterococcus*, *Streptococcus*, *Lactobacillus*, *Escherichia*, and *Klebsiella* (**Fig E**) as annotated by Greengenes and NCBI databases. At all four institutions, the cumulative incidence of intestinal domination by any organism was >50% by day 0 and was >87% by day +28 (**Fig F**). We performed additional analyses in the largest cohort and found that the low-diversity state is associated with exposure to broad-spectrum antibiotics, conditioning intensity, and low calorie intake.

We demonstrate that HCT patients at 4 institutions on 3 continents presented with microbiota configurations that were similar to one another and distinct from those of healthy individuals. Severe microbiota injury as revealed by domination is a common event whose development begins before allograft infusion, and pre-HCT microbiota injury predicts poor overall survival. These observations suggest the pre-HCT period as a window of opportunity to (a) assess microbiota injury as part of comorbidity evaluation, (b) inform selection of antibiotic prophylaxis, gut-decontamination, GVHD-prophylaxis, or conditioning regimens, and (c) intervene with microbiota injury-remediation or prevention strategies.

Figure.

Disclosures

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